



ASCHAM SCHOOL
MATHEMATICS EXTENSION 1
TRIAL EXAMINATION

2006

Time : 2 hours + 5 minutes
 reading time

Instructions:

Attempt all questions

All questions are of equal value

All necessary working should be shown for every question.

Full marks may not be awarded for careless or badly arranged work

A Table of Standard Integrals is provided

Approved calculators may be used

Each question should be answered in a separate booklet

Do not use whiteout, part marks may be awarded for scored out work if it is legible

Question 1 (12 marks)

- (a) Find $\int \frac{1}{\sqrt{4-x^2}} dx$ [1]
- (b) Sketch the region in the number plane defined by $y > |x| - 1$ [2]
- (c) Find the domain and range of $y = \sqrt{x^3 - 9}$ [2]
- (d) Find $\lim_{x \rightarrow 0} \frac{x}{\sin 2x}$ [2]
- (e) The parametric equation of a function is
 $x = 2t^2, y = 4 - t$
 Find the cartesian equation. [1]
- (f) A $(x, 10)$ and B $(6, y)$. The point P $(5, 4)$ divides AB externally in the ratio 3:1. Find x and y [2]
- (g) Find $\frac{d}{d\theta}(\cos^3 2\theta)$ [2]

Question 2 (12 marks)

Begin a new booklet

- (a) (i) Show $\frac{d}{dx}(x\sqrt{1-x^2} + \sin^{-1} x) = 2\sqrt{1-x^2}$ [2]
- (ii) Hence evaluate $\int_0^{\frac{1}{2}} \sqrt{1-x^2} dx$ [2]
- (b) Using the substitution $u = \log_e x$, evaluate $\int_e^{e^2} \frac{1}{x \log_e x} dx$ [3]
- (c) The polynomial equation $3x^3 - 2x^2 + 3x - 4 = 0$ has roots α, β, λ .
 Find the exact value of $\frac{1}{\alpha\beta} + \frac{1}{\alpha\lambda} + \frac{1}{\beta\lambda}$ [2]
- (d) Consider the polynomial $P(x) = x^3 + ax^2 + bx + 2$ which has factors $x+1$ and $x-2$. Find the values of a and b . [3]

Question 3 (12 marks)

Begin a new booklet

- (a) Find the general solution of $\cos x \cos 27^\circ + \sin x \sin 27^\circ = \cos 2x$ [3]
- (b) Drinks for a barbeque have been left in the sun and their temperature has risen to 30°C . They are placed in the freezer where the temperature is maintained at -5°C . After t minutes, the temperature $T^\circ\text{C}$ of the drinks is changing so that $\frac{dT}{dt} = -k(T + 5)$
- (i) Prove that $T = Ae^{-kt} - 5$ satisfies the differential equation, and find the value of A . [2]
- (ii) After 20 minutes the temperature of the drinks has fallen to 20°C . How long after they are put in the fridge will it take before the drinks begin to freeze? Assume that freezing point is 0°C . [2]
- (c) (i) Prove using calculus that the equation $x^3 + 2x + 4 = 0$ has only one real root α [2]
- (ii) Show that $-2 < \alpha < -1$ [1]
- (iii) Starting with an initial approximation $\alpha = -1$, use one application of Newton's method to find a further approximation for α . [2]

Question 4 (12 marks)

Begin a new booklet

- (a) A particle is moving along the x-axis. Its speed v m/s at position x metres is given by
- $$v = \sqrt{5x - x^2}$$
- Find the acceleration when $x = 2$. [2]

- (b) A particle moves along the x-axis according to the equation

$$x = \cos 2t - \sqrt{3} \sin 2t$$

where x metres is the displacement after t seconds from the origin O .

- (i) Express x in the form $R \cos(2t + \alpha)$ where $R > 0$ and $0 \leq \alpha \leq \frac{\pi}{2}$ [2]
- (ii) Prove that the particle moves in simple harmonic motion. [2]
- (iii) Find the amplitude and period of the motion. [2]
- (iv) Determine whether the particle is initially moving towards O or away from O , and whether it is initially speeding up or slowing down. Justify your answers. [2]
- (v) Find the time at which the particle first returns to its starting point. [2]

Question 5 (12 marks)

Begin a new booklet

- (a) (i) From a lighthouse L , the bearing of ships A and B are 035° and 145° respectively. Show this on a diagram and find $\angle ALB$. [1]
- (ii) Lighthouse LT is 120 metres high. The angle of elevations from ships A and B to the top of the lighthouse are 40° and 50° respectively. Find the distance between the ships. [3]
- (b) (i) Show that $f(x) = \sin^{-1}(\cos x)$ is an even function. [1]
- (ii) Differentiate $f(x) = \sin^{-1}(\cos x)$ and hence find the gradient for $0 < x < \pi$. [2]
- (iv) Evaluate $f(0)$, $f(-\pi)$ and $f(\pi)$ [1]
- (iii) Sketch $f(x)$ for $-\pi \leq x \leq \pi$ [1]
- (c) Use mathematical induction to prove that
- $$1^3 + 2^3 + 3^3 + \dots + n^3 = \frac{n^2(n+1)^2}{4} \quad \text{for integer } n \geq 1$$
- [3]

Question 6 (12 marks)

Begin a new booklet

(a) Given that $\sin^{-1} x$ and $\cos^{-1} x$ are acute,

(i) Show that $\sin(\sin^{-1} x - \cos^{-1} x) = 2x^2 - 1$ [2]

(ii) Solve the equation $\sin^{-1} x - \cos^{-1} x = \sin^{-1}(0.5)$ [2]

(b) A particle is projected from a point O with velocity V m/s at an angle θ above the horizontal. At time t seconds it has horizontal and vertical components x metres and y metres respectively from O. The acceleration due to gravity is g m/s².(i) Given the equations below, derive equations for horizontal displacement x and vertical displacement y [2]

$$\ddot{x} = 0, \quad \ddot{y} = -g$$

$$\dot{x} = V \cos \theta, \quad \dot{y} = V \sin \theta - gt$$

(ii) Hence show that the equation of the path is

$$y = x \tan \theta - \frac{gx^2}{2V^2} (1 + \tan^2 \theta)$$
 [2]

(c) A particle is projected from O with velocity 60 m/s at an angle α above the horizontal. T seconds later, another particle is also projected from O with velocity 60 m/s at an angle β above the horizontal, where $\beta < \alpha$. The two particles collide 240 metres horizontally and 80 metres vertically from O. Taking $g = 10 \text{ m/s}^2$, and using the results from (b):

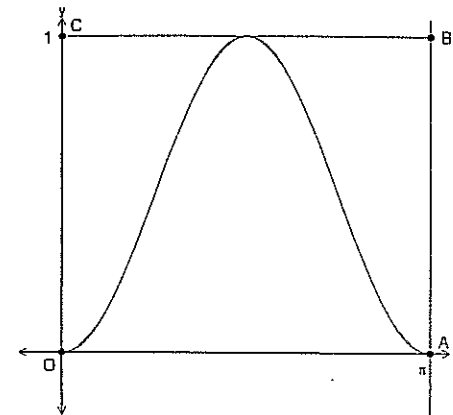
(i) Show that $\tan \alpha = 2$ and $\tan \beta = 1$ [2]

(ii) Find the value of T in simplest exact form. [2]

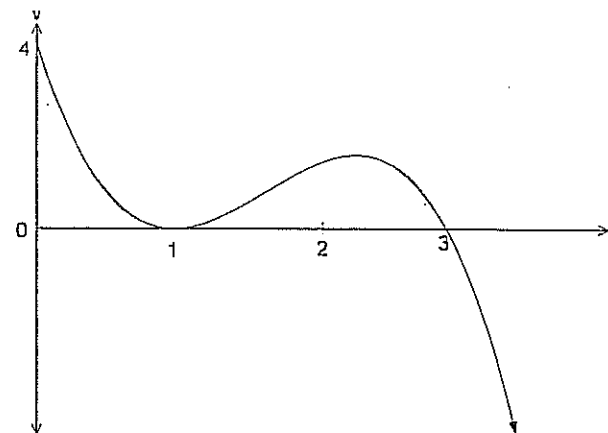
Question 7 (12 marks)

Begin a new booklet

(a)

The rectangle OABC has vertices O (0, 0), A (π , 0), B (π , 1), C (0, 1).The curve $y = \sin^2 x$ is shown. Use calculus methods to show that the area under the curve is half the area of rectangle OABC [3]

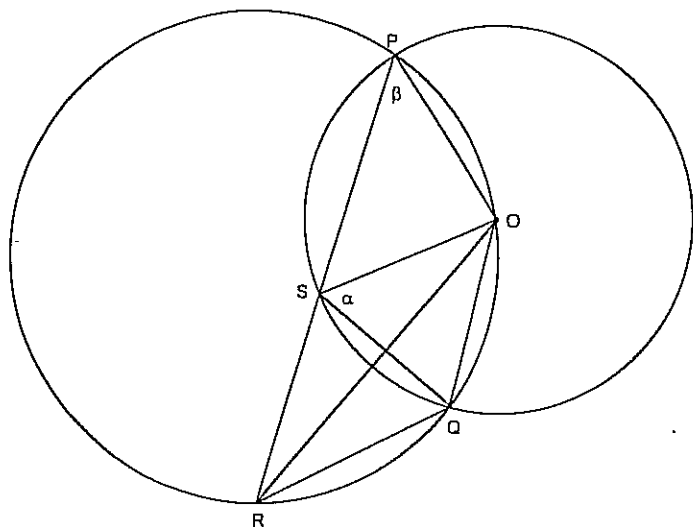
(b) A particle P moves along a straight line. A velocity-time graph for P is shown below.



(i) Between what times does the particle travel to the right? [1]

(ii) Sketch a displacement-time graph for P given that the particle starts 2 metres to the left of O. [2]

(c)



O is a point on the larger circle. The smaller circle has centre O. The circles intersect at P and Q. PR is a chord of the larger circle that cuts the smaller circle at S.

Copy the diagram into your answer booklet (about half a page)

Let $\angle SPO = \beta$, $\angle OSQ = \alpha$

- | | | |
|-------|--|-----|
| (i) | Explain why $\angle PSO = \beta$ | [1] |
| (ii) | Prove that $\angle SQR = 180 - (\alpha + \beta)$ | [2] |
| (iii) | Prove that $SQ \perp OR$ | [3] |

End of Examination

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①

338 / a) $\int \frac{1}{\sqrt{4-x^2}} dx = \sin^{-1} \frac{x}{2} + C$

①

b)



②

c) $x^2 - 9 \geq 0$

D: $x \leq -3$ or $x \geq 3$. ✓

②

R: $y \geq 0$. ✓

d) $\lim_{x \rightarrow 0} \frac{x}{\sin 2x} = \lim_{2x \rightarrow 0} \frac{2x}{\sin 2x} \cdot \frac{1}{2}$
 $= \frac{1}{2} \cdot 1 = \frac{1}{2}$. ✓

②

e) $x = 2t^2$ $y = 4 - t$
 $t = 4 - y$
 $\therefore x = 2(4 - y)^2$. ✓

①

f) A(x, 10) B(6, y)
 $-3 : 1$

$5 = \frac{-3 \times 6 + x}{-3 + 1} \sqrt{2}$ $4 = \frac{-3 \times y + 10}{-3 + 1} \sqrt{2}$

②

$-10 = -18 + x$ $-8 = -3y + 10$
 $x = 8 \sqrt{2}$ $3y = 18$
 $y = 6 \sqrt{2}$

g) $\frac{d}{d\theta} (\cos^3 2\theta) = 3 \cos^2 2\theta (-\sin 2\theta) \cdot 2$
 $= -6 \cos^2 2\theta \sin 2\theta$

②

(2)

$$2 a) \frac{d}{dx} (x\sqrt{1-x^2} + \sin^{-1} x) \quad \checkmark$$

$$(i) = x^{\frac{1}{2}}(1-x^2)^{-\frac{1}{2}}(-2x) + \sqrt{1-x^2} \cdot 1 + \frac{1}{\sqrt{1-x^2}} \quad \checkmark$$

$$= \frac{-x^2}{\sqrt{1-x^2}} + \sqrt{1-x^2} + \frac{1}{\sqrt{1-x^2}}$$

$$= \frac{-x^2 + 1 - x^2 + 1}{\sqrt{1-x^2}}$$

(2)

$$= \frac{2(1-x^2)}{\sqrt{1-x^2}} \quad \checkmark$$

$$= 2\sqrt{1-x^2}$$

$$(ii) \int_0^{\frac{1}{2}} \sqrt{1-x^2} dx = \frac{1}{2} \left[x\sqrt{1-x^2} + \sin^{-1} x \right]_0^{\frac{1}{2}} \quad \checkmark$$

$$= \frac{1}{2} \left[\left(\frac{1}{2} \sqrt{\frac{3}{4}} + \sin^{-1} \frac{1}{2} \right) - 0 \right]$$

$$= \frac{1}{2} \left[\frac{\sqrt{3}}{2} + \frac{\pi}{6} \right] \quad \checkmark$$

$$= \frac{1}{12} [3\sqrt{3} + \pi]$$

(2)

$$(b) \int_e^{e^2} \frac{1}{x \log x} dx \quad \text{let } u = \log_e x$$

$$du = \frac{dx}{x} \quad \checkmark$$

$$= \int_1^2 \frac{1}{u} du \quad \checkmark$$

$$= [\ln u]_1^2$$

$$= \ln 2 - \ln 1$$

$$= \ln 2 \quad \checkmark$$

Q2 cont.

(3)

$$(c) 3x^3 - 2x^2 + 3x - 4 = 0$$

$$\text{roots } \alpha, \beta, \gamma.$$

$$\frac{1}{\alpha\beta} + \frac{1}{\alpha\gamma} + \frac{1}{\beta\gamma}$$

(2)

$$= \frac{\gamma + \beta + \alpha}{\alpha\beta\gamma} \quad \checkmark$$

$$= \frac{2/3}{4/3} \quad \checkmark$$

$$= \frac{1}{2} \quad \checkmark$$

$$(d) P(x) = x^3 + ax^2 + bx + 2$$

$$(x+1); x-2 \text{ are factors}$$

$$\therefore P(-1) = 0 \quad \checkmark$$

$$P(2) = 0$$

$$-1 + a - b + 2 = 0$$

$$a - b = -1 \quad (1)$$

$$8 + 4a + 2b + 2 = 0$$

$$2a + b = -3 \quad (2)$$

(3)

$$(1) + (2)$$

$$3a = -4$$

$$a = -4/3 \quad \checkmark$$

$$\rightarrow (1)$$

$$-4/3 - b = -1$$

$$b = 1 - 4/3$$

$$= -1/3 \quad \checkmark$$

④

3 a) $\cos x \cos 27^\circ + \sin x \sin 27^\circ = \cos 2x$
 $\cos(x - 27^\circ) = \cos 2x$ ✓
 $x - 27^\circ = \pm 2x + 360n \quad \frac{1}{2}, \quad n \in \{\text{integers}\} \quad \frac{1}{2}$
 $x = (27 + 360n)^\circ \quad \text{or} \quad x = \frac{27 + 360n}{3}$
 $x = (360n - 27)^\circ \quad \frac{1}{2} \quad \text{or} \quad x = (9 + 120n)^\circ \quad \frac{1}{2}$

b) $\frac{dT}{dt} = -k(T+5)$
 when $t=0, T=30$
 (i) $T = Ae^{-kt} - 5 \quad \text{or} \quad Ae^{-kt} = T+5 \quad \text{①}$
 $\frac{dT}{dt} = A \cdot (-k) e^{-kt}$
 $= -k Ae^{-kt} \quad \text{sub in ①.} \quad \checkmark$
 $= -k(T+5)$

② $T = Ae^{-kt} - 5$ is a soln of $\frac{dT}{dt} = -k(T+5)$
 when $t=0, T=30$
 $\therefore 30 = Ae^0 - 5$
 $A = 35. \quad \checkmark$

(ii) when $t=20, T=20$
 $\therefore 20 = 35e^{-k \times 20} - 5$
 $\frac{25}{35} = e^{-20k}$
 $-20k = \ln \frac{5}{7}$
 $k = -\frac{1}{20} \ln \frac{5}{7} \quad \checkmark$
 $(= \frac{1}{20} \ln \frac{7}{5})$

② when $T=0$
 $0 = 35e^{\frac{1}{20} \ln \frac{5}{7} \cdot t} - 5$
 $\frac{5}{35} = e^{\frac{1}{20} \ln \frac{5}{7} \cdot t}$

⑤

$t = \frac{20 \ln \frac{7}{5}}{\ln \frac{5}{7}} \quad \checkmark$
 $= 115.66. \quad \checkmark$

$\therefore$ After about 116 mins the drinks begin to freeze.

(c)
 (i) let $f(x) = x^3 + 2x + 4$
 $f'(x) = 3x^2 + 2 > 0$ for all x ✓
 $\therefore f(x)$ increases for all x ✓
 $\therefore x^3 + 2x + 4 = 0$ has only 1 root α .

(ii) $f(-2) = (-2)^3 + 2 \times (-2) + 4$
 $= -8$
 $< 0 \quad \checkmark$

$f(-1) = (-1)^3 + 2 \times (-1) + 4$
 $= 1$
 $> 0 \quad \checkmark$

$f(\alpha) = 0$
 $f(-2) < f(\alpha) < f(-1)$
 $-2 < \alpha < -1$ since $f(x)$ is increasing

(iii) $\alpha_1 = x_0 - \frac{f(x_0)}{f'(x_0)} \quad \checkmark$ let $x_0 = -1$
 $= -1 - \frac{1}{5}$
 $= -\frac{6}{5} \quad \checkmark$
 $f'(-1) = 3 \times (-1)^2 + 2 = 5$

② $\therefore$ A further approx. for α is $-\frac{6}{5}$.

Q4. a)

$$V = \sqrt{5x - x^2}$$

$$\ddot{x} = \frac{d}{dx} \left(\frac{1}{2} V^2 \right) \downarrow$$

$$= \frac{d}{dx} \left(\frac{1}{2} (5x - x^2) \right)$$

$$= \frac{1}{2} (5 - 2x) \quad \checkmark$$

$$= \frac{5}{2} - x.$$

$$= \frac{5}{2} - 2 \text{ when } x=2 \quad \checkmark$$

$\therefore$ When $x=2$, acceleration is $\frac{1}{2} \text{ m/s}^2$. $\checkmark$

b).

$$x = \cos 2t - \sqrt{3} \sin 2t.$$

$$(i) R \cos(2t + \alpha) = R \cos 2t \cos \alpha - R \sin 2t \sin \alpha$$

$$\text{and } x = \cos 2t - \sqrt{3} \sin 2t.$$

Equating coef:

$$R \cos \alpha = 1 \quad \checkmark$$

$$R \sin \alpha = \sqrt{3}$$

$$\text{Dividing } \tan \alpha = \sqrt{3}$$

$$\alpha = 60^\circ \quad \checkmark$$

(2)

Squaring & adding

$$R^2 (\sin^2 \alpha + \cos^2 \alpha) = \sqrt{3}^2 + 1^2$$

$$R^2 = 4$$

$$R = 2 \quad (R > 0) \quad \checkmark$$

$$\therefore x = 2 \cos \left(2t + \frac{\pi}{3} \right).$$

(ii)

$$\dot{x} = -2 \sin \left(2t + \frac{\pi}{3} \right) \times 2$$

$$= -4 \sin \left(2t + \frac{\pi}{3} \right) \quad \checkmark$$

$$\ddot{x} = -8 \cos \left(2t + \frac{\pi}{3} \right)$$

$$= -4x$$

$\therefore$ Particle is in SHM since in form

$$\ddot{x} = -k^2 x.$$

(6)

(iii)

$$\text{Amplitude} = 2 \quad \checkmark \quad \left(\text{from } x = 2 \cos \left(2t + \frac{\pi}{3} \right) \right)$$

$$-k^2 = 4$$

$$k = 2 \quad (k > 0)$$

$$\therefore \text{Period} = \frac{2\pi}{k} \\ = \pi. \quad \checkmark$$

(2)

(iv)

$$\text{When } t=0, \quad x = 2 \cos \frac{\pi}{3}$$

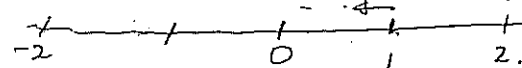
$$= 1 \quad \checkmark$$

$\therefore$ Unit to right of 0.

$$\dot{x} = -4 \sin \frac{\pi}{3}$$

$$= -2\sqrt{3}$$

$\therefore$ moving $\leftarrow$ $\checkmark$



$\therefore$ Initially particle moves towards 0. $\checkmark$

(2)

$$\text{When } t=0 \quad \ddot{x} = -4x$$

$$= -4$$

$\therefore$ force acts $\leftarrow$ $\checkmark$

$\therefore$ Particle is speeding up, when $t=0$. $\checkmark$

(v)

Returns to its starting point when $x=1$

$$2 \cos \left(2t + \frac{\pi}{3} \right) = 1 \quad \checkmark$$

$$\cos \left(2t + \frac{\pi}{3} \right) = \frac{1}{2}$$

$$2t + \frac{\pi}{3} = \frac{\pi}{3}, \frac{5\pi}{3} \quad \checkmark$$

$$2t = 0, \quad 4\pi/3$$

$$t = 0, \quad 2\pi/3 \quad \checkmark$$

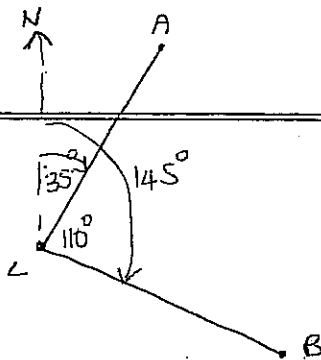
$\therefore$ Returns to start after $\frac{2\pi}{3}$ secs $\checkmark$

(2)

(7)

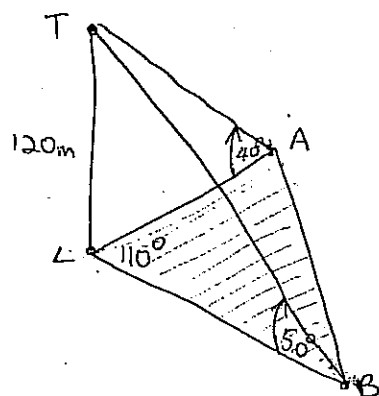
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Q5a)
(i)



$$\angle ALB = 110^\circ$$

(ii)



$$AL = 120 \cot 40^\circ$$

$$BL = 120 \cot 50^\circ$$

$$AB^2 = AL^2 + BL^2 - 2AL \cdot BL \cos \angle ALB$$

$$= (120 \cot 40^\circ)^2 + (120 \cot 50^\circ)^2 - 2 \times 120 \cot 40^\circ \times 120 \cot 50^\circ \times \cos 110^\circ$$

$$= 120^2 [\cot^2 40^\circ + \cot^2 50^\circ - 2 \cot 40^\circ \cot 50^\circ \cos 110^\circ]$$

$$= 40441.033 \dots$$

$$AB = 201.099$$

$\therefore$ Distance between the ships is 201m (to n.m).

0) (i)

$$f(x) = \sin^{-1}(\cos x)$$

$$f(-x) = \sin^{-1}(\cos(-x))$$

$$= \sin^{-1}(\cos x)$$

$$= f(x)$$

9

(ii)

$$f(x) = \sin^{-1}(\cos x)$$

$$f'(x) = \frac{1}{\sqrt{1 - \cos^2 x}} \times -\sin x$$

$$= \frac{-\sin x}{\sqrt{\sin^2 x}}$$

$$= -\frac{\sin x}{\sin x}$$

$$= -1$$

$\therefore$ Gradient = -1 for $0 < x < \pi$ where $\sin x > 0$

(iii)

$$f(0) = \sin^{-1}(\cos 0)$$

$$= \sin^{-1} 1$$

$$= \frac{\pi}{2}$$

$$f(-\pi) = \sin^{-1}(\cos(-\pi))$$

$$= \sin^{-1}(-1)$$

$$= -\frac{\pi}{2}$$

(i)

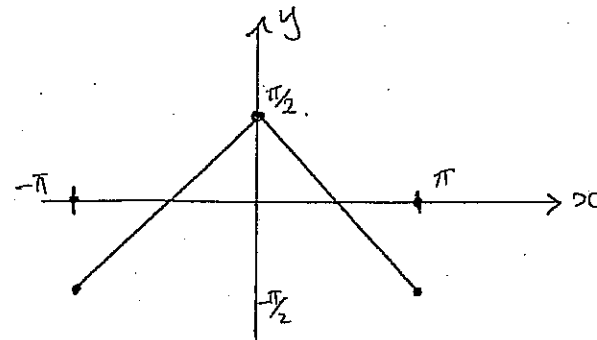
$$f(\pi) = \sin^{-1}(\cos \pi)$$

$$= \sin^{-1}(-1)$$

$$= -\frac{\pi}{2}$$

(iv)

(i)



5 cont.

(10)

(10) Let $P(n)$ be $1^3 + 2^3 + 3^3 + \dots + n^3 = \frac{n^2(n+1)^2}{4}$ integers $n \geq 1$

$P(1)$ is that $1^3 = \frac{1^2(1+1)^2}{4} = 1$
 $1^3 = 1 \therefore P(1)$ is true ✓

Assume $P(k)$: $1^3 + 2^3 + \dots + k^3 = \frac{k^2(k+1)^2}{4}$ is true ✓

and Prove $P(k+1)$ is true
 i.e. Prove $1^3 + 2^3 + \dots + (k+1)^3 = \frac{(k+1)^2(k+1+1)^2}{4}$

$$\text{LHS} = 1^3 + 2^3 + \dots + k^3 + (k+1)^3$$

$$= \frac{k^2(k+1)^2}{4} + (k+1)^3$$

$$= (k+1)^2 \left[\frac{k^2}{4} + k+1 \right]$$

$$= \frac{(k+1)^2}{4} [k^2 + 4k + 4]$$

$$= \frac{(k+1)^2 (k+2)^2}{4}$$

= RHS

$\therefore$ If $P(k)$ is true then $P(k+1)$ is true
 Since $P(1)$ is true the result is proved by Mathematical induction

(11)

Q6(a) $\sin^{-1}x$ and $\cos^{-1}x$ are acute angles.

(i) $\sin(\sin^{-1}x - \cos^{-1}x) = \sin(A-B)$ where $\sin A = x$

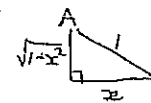
$\cos B = x$

$$= \sin A \cos B - \cos A \sin B$$

$$= x \cdot x - \sqrt{1-x^2} \cdot \sqrt{1-x^2}$$

$$= x^2 - (1-x^2)$$

$$= 2x^2 - 1$$



(2)

(ii) $\sin^{-1}x - \cos^{-1}x = \sin^{-1}(0.5)$

taking sine of both sides

$$\sin(\sin^{-1}x - \cos^{-1}x) = \sin(\sin^{-1}0.5)$$

$$2x^2 - 1 = 0.5$$

$$2x^2 = 1.5$$

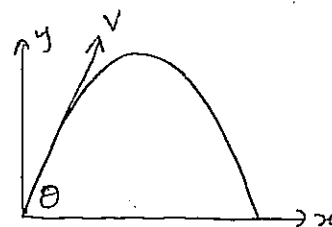
$$x^2 = \frac{3}{4}$$

$$x = \pm \frac{\sqrt{3}}{2}$$

$$= \frac{\sqrt{3}}{2} \text{ since } \sin^{-1}x \text{ is acute}$$

(2)

(b)



$$\ddot{x} = 0 \quad \ddot{y} = -g$$

$$\dot{x} = V \cos \theta \quad y = V \sin \theta - gt$$

$$(i) \quad x = \int V \cos \theta dt$$

$$= Vt \cos \theta + c_1$$

when $t=0, x=0 \Rightarrow c_1=0$

$$x = Vt \cos \theta \quad \checkmark \text{ (1)}$$

(2)

$$y = \int V \sin \theta - gt dt$$

$$= Vt \sin \theta - \frac{g}{2} t^2 + c_2$$

when $t=0, y=0 \Rightarrow c_2=0$

$$y = Vt \sin \theta - \frac{g}{2} t^2 \quad \checkmark \text{ (2)}$$

(12)

(ii) from ① $t = \frac{x}{v \cos \theta}$ ✓

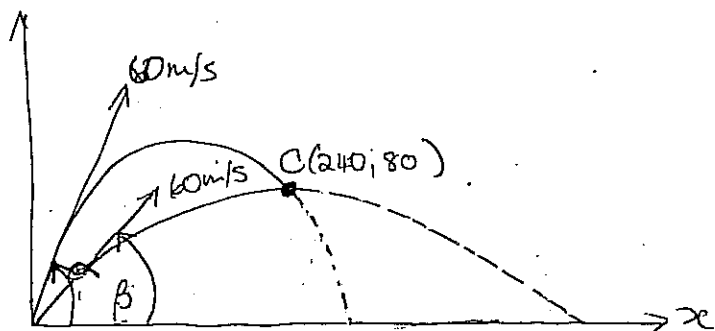
→ ②

$$y = \frac{\sqrt{x}}{\sqrt{\cos \theta}} \sin \theta - \frac{g}{2} \frac{x^2}{v^2 \cos^2 \theta} \quad \checkmark$$

$$y = x \tan \theta - \frac{gx^2}{2v^2} \sec^2 \theta \quad \checkmark$$

$$y = x \tan \theta - \frac{gx^2}{2v^2} (1 + \tan^2 \theta) \quad \text{③}$$

(c)



Collide at point C(240, 80).

(i) For 1st particle which passes through $x=240$

→ ③

$$80 = 240 \tan \alpha - \frac{10 \times 240^2}{2 \times 60^2} (1 + \tan^2 \alpha) \quad \checkmark$$

÷ 80

$$1 = 3 \tan \alpha - (1 + \tan^2 \alpha)$$

$$\tan^2 \alpha - 3 \tan \alpha + 2 = 0$$

$$(\tan \alpha - 2)(\tan \alpha - 1) = 0$$

$$\therefore \tan \alpha = 2 \text{ or } 1 \quad \checkmark$$

Similarly for the second particle

$$\tan \beta = 2 \text{ or } 1$$

$$\text{Since } \beta < \alpha \quad \tan \alpha = 2 \quad \checkmark$$

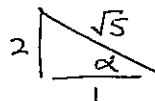
$$\tan \beta = 1 \quad \checkmark$$

(13)

(ii) find T, the time between projections when $x=240$

$$240 = 60 t \cos \alpha \quad \checkmark \text{ for 1st particle}$$

$$240 = 60 t \cdot \frac{1}{\sqrt{5}}$$



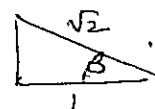
$$t = \frac{240\sqrt{5}}{60} = 4\sqrt{5} \quad \checkmark$$

②

$$240 = 60 t \cos \beta \text{ for 2nd particle}$$

$$t = \frac{240}{60} \cdot \sqrt{2}$$

$$= 4\sqrt{2} \quad \checkmark$$



$$\therefore T = 4\sqrt{5} - 4\sqrt{2} = 4(\sqrt{5} - \sqrt{2}) \text{ secs.} \quad \checkmark$$

7a)

Area under curve

$$= \int_0^{\pi} \sin^2 x \, dx$$

$$\cos 2x = 1 - 2\sin^2 x$$

$$2\sin^2 x = 1 - \cos 2x$$

$$= \frac{1}{2} \int_0^{\pi} (1 - \cos 2x) \, dx \quad \checkmark$$

$$= \frac{1}{2} \left[x - \frac{\sin 2x}{2} \right]_0^{\pi}$$

$$= \frac{1}{2} [(\pi - 0) - (0 - 0)]$$

$$= \frac{\pi}{2} \quad \checkmark$$

③

Area of rectangle OABC = $l \times b$

$$= \pi \times 1$$

$$= \pi \quad \checkmark$$

 $\therefore$ Area under curve is half the area of rectang

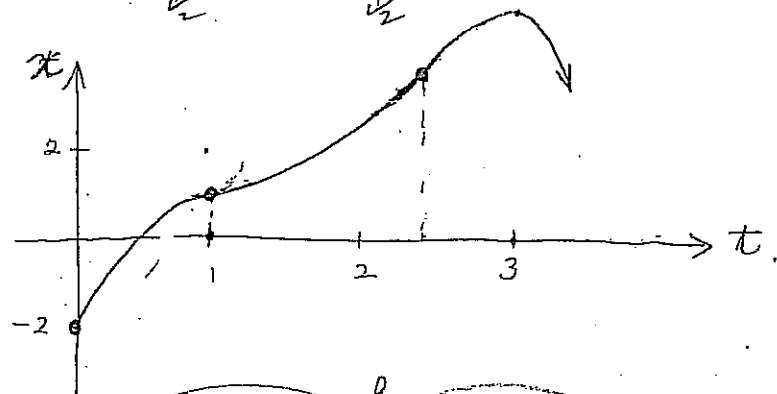
(14)

- (b) (i) Particle travels to the right when $v > 0$.
 $0 \leq t < 3$, $t \neq 1$
 $\swarrow$ $\searrow$

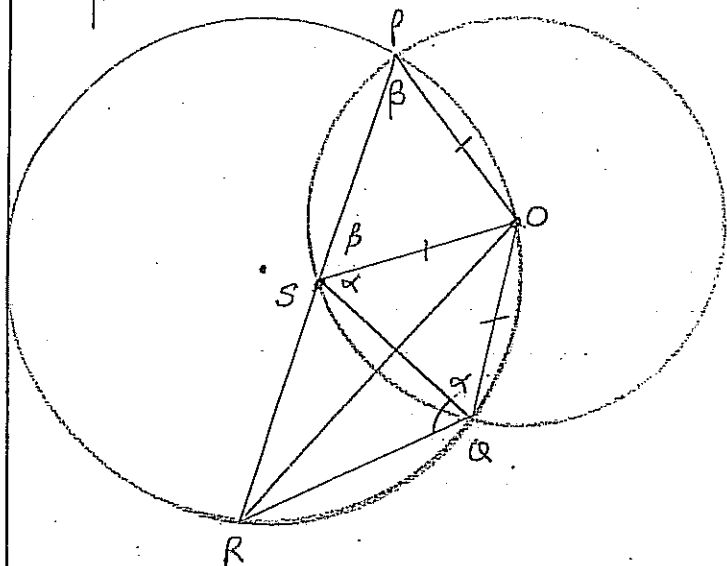
①

(ii)

②



(c)



- (i) $\angle SPO = \beta$ $OQ = OP = OS$ (radii) ✓
 $\therefore \angle PSO = \beta$ (LS opp equal sides in $\triangle POS$) ✓
 (ii) $\angle OSQ = \angle OQS = \alpha$ (base LS of $\triangle OSQ$, $OS = OQ$ radii) ✓
 $\angle SPO + \angle OQR = 180$ (opp LS of cyclic quad $POQR$) ✓
 $\beta + \angle OSR + \alpha = 180$
 $\angle OSR = 180 - (\alpha + \beta)$
 (iii) $\angle RSQ = 180 - (\alpha + \beta)$ (straight L at S) ✓

(15)

③

- $\therefore \angle RSQ = \angle OSR = 180 - (\alpha + \beta)$.
 $\therefore RS = RQ$ (sides opp = LS in $\triangle RSQ$) ✓
 $\therefore OSRS$ is a kite (2 prs adj sides =)
 $\therefore SQ \perp OR$ (diagonals of a kite intersect at right LS. ✓)